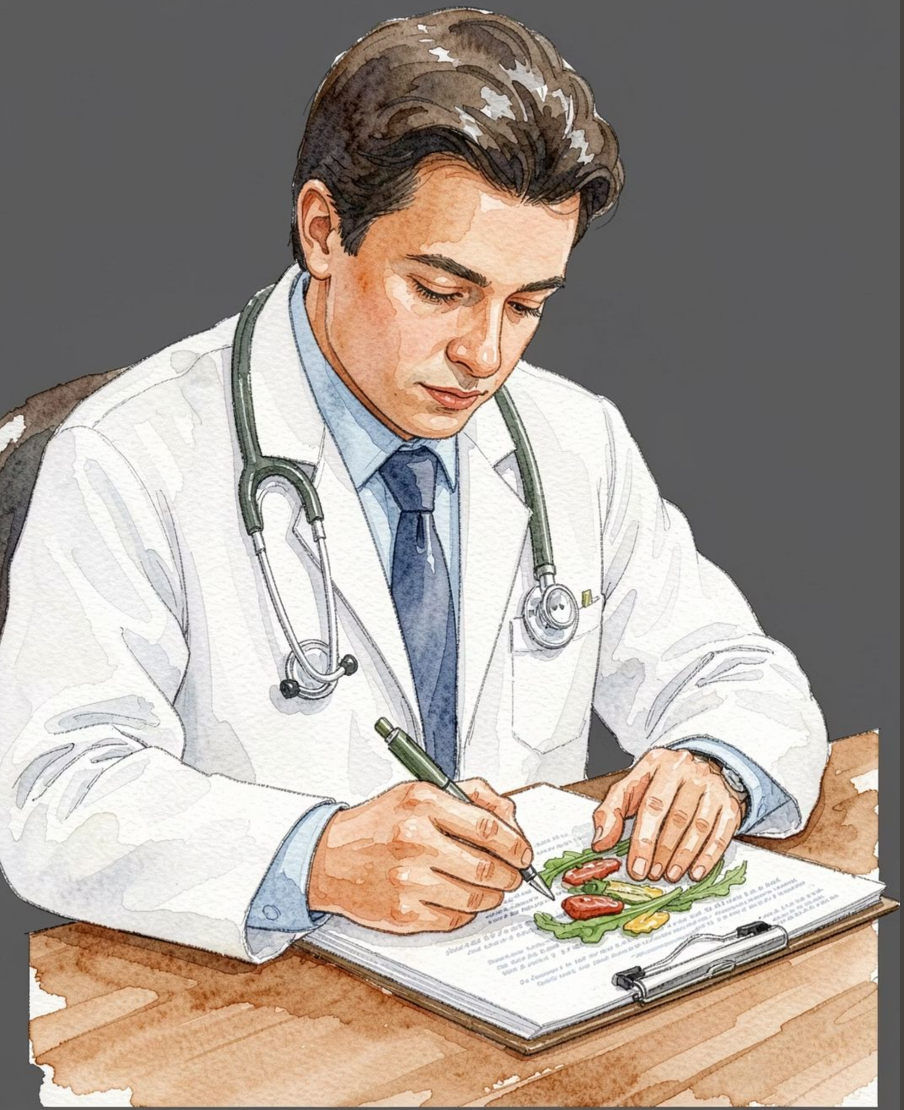


Medical Insurance Cost Prediction using Linear Regression

Project Documentation — Presented by Shafaat

Problem Statement

The Challenge

Medical insurance costs vary widely — hard to estimate without a data-driven approach

The Goal

Predict insurance charges based on demographic and health-related features

The Solution

Build a supervised Machine Learning model using Linear Regression

Objectives

Explore & Preprocess

Explore and preprocess the medical insurance dataset

Build & Train

Build and train a Linear Regression model

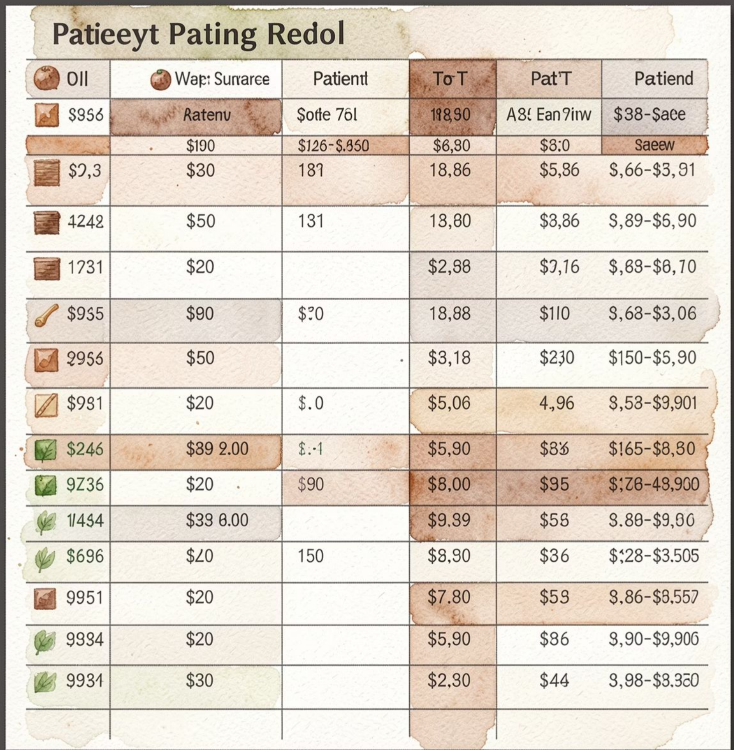
Evaluate Performance

Evaluate model performance using standard regression metrics

Establish Baseline

Use the model as a reliable baseline for insurance cost prediction

Dataset Overview



Source

Medical Insurance dataset — a structured, publicly available dataset for regression analysis.

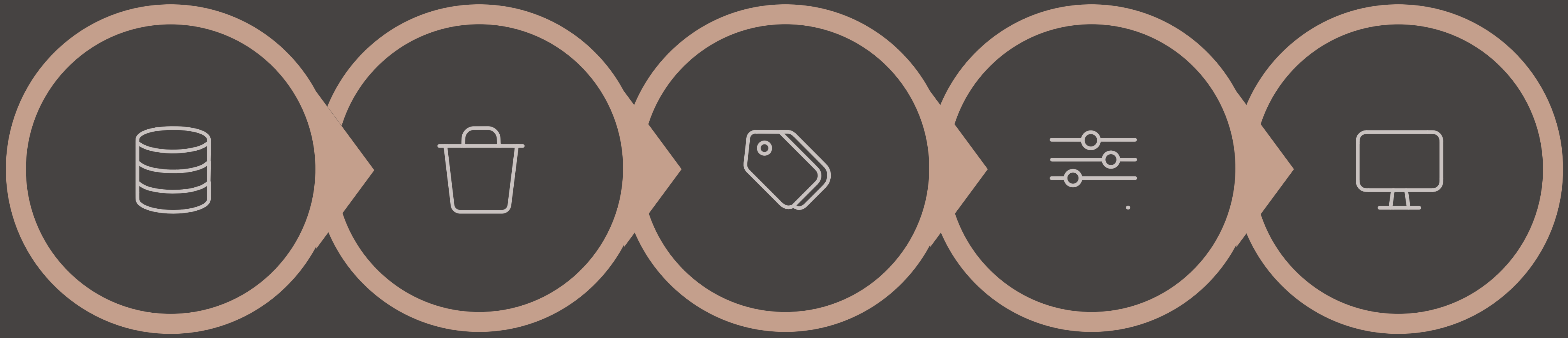
Input Features

- **Age** — policyholder's age
- **Sex** — gender of the insured
- **BMI** — body mass index
- **Children** — number of dependents
- **Smoker** — smoking status
- **Region** — geographic region

Target Variable

Charges — insurance cost in USD

Data Preprocessing



Handle Missing

Remove Duplicates

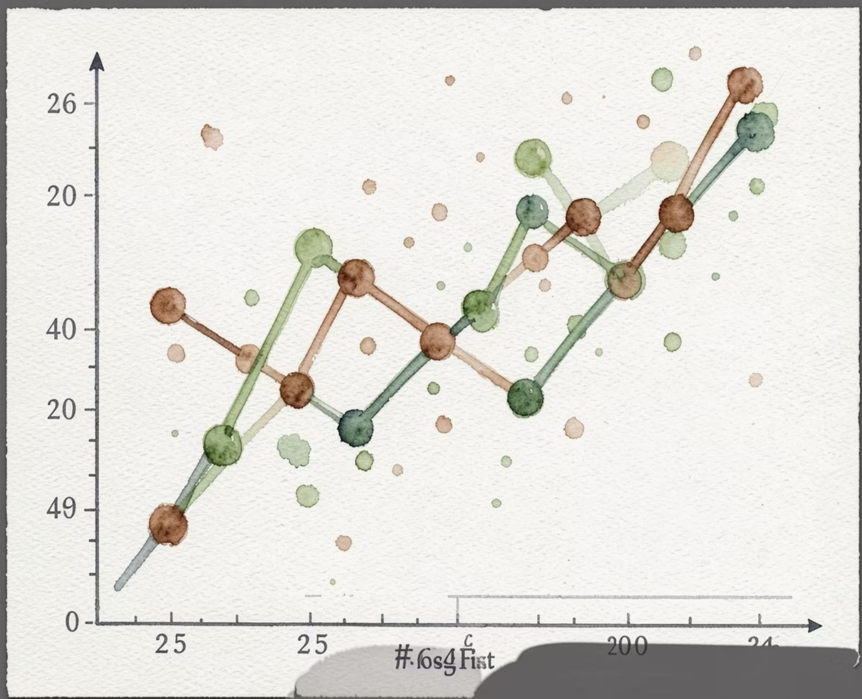
Encode Categorical

Scale Features

Train/Test Split

Each step was carefully applied to ensure data quality and model readiness — from cleaning raw records to preparing a balanced train/test split for reliable evaluation.

Model: Linear Regression



How It Works

Linear Regression models the relationship between input features and the target variable (charges) by learning a set of coefficients for each feature.

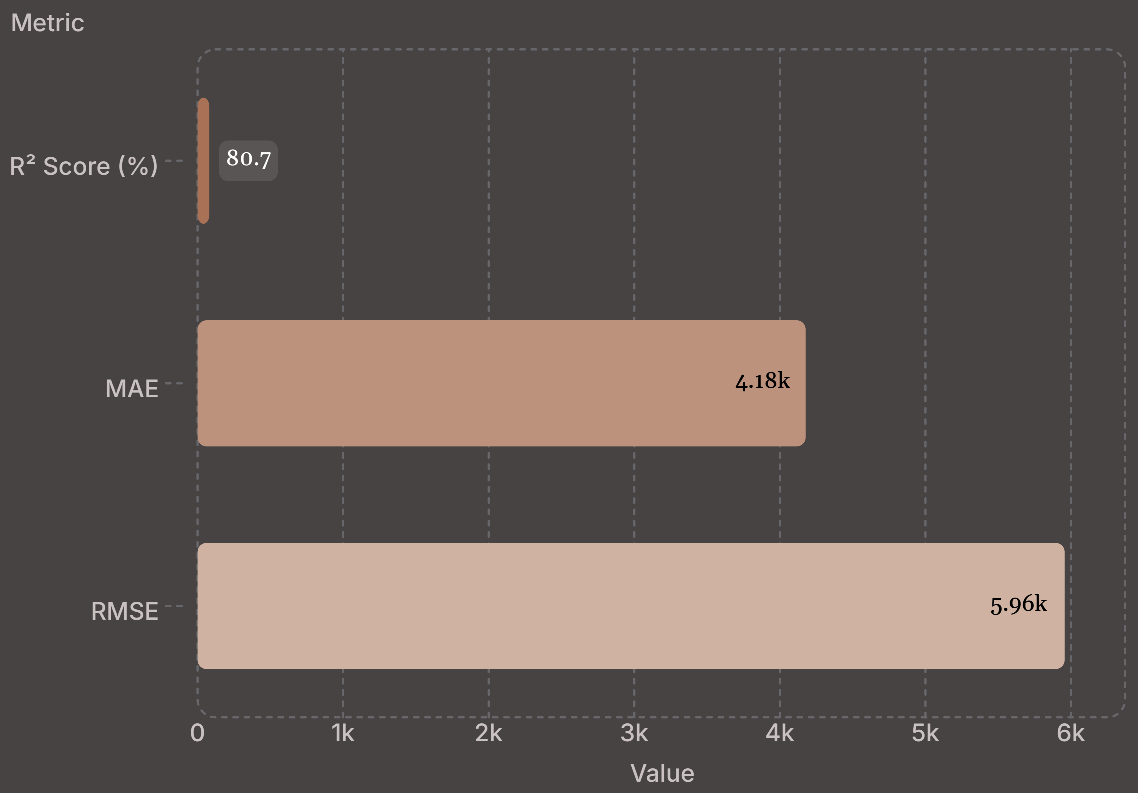
Optimization

Finds the best-fit line by **minimizing the sum of squared errors** between predicted and actual values.

Why Linear Regression?

Simple, interpretable, and effective for continuous value prediction — making it an ideal baseline model.

Model Evaluation Results



R^2 Score: 0.8069

Model explains **80.69%** of the variance in insurance charges — strong predictive fit

MAE: 4,177.05

Average absolute prediction error in USD

RMSE: 5,956.34

Standard deviation of prediction errors — penalizes larger deviations

Conclusion & Future Work

Key Takeaway

The Linear Regression model achieved **strong predictive performance** with an R^2 of 80.69%, establishing a solid baseline for medical insurance cost estimation.

Future Improvements

- Explore **Random Forest** and **XGBoost** for higher accuracy
- Apply feature engineering to capture non-linear relationships
- Expand dataset for better generalization across populations